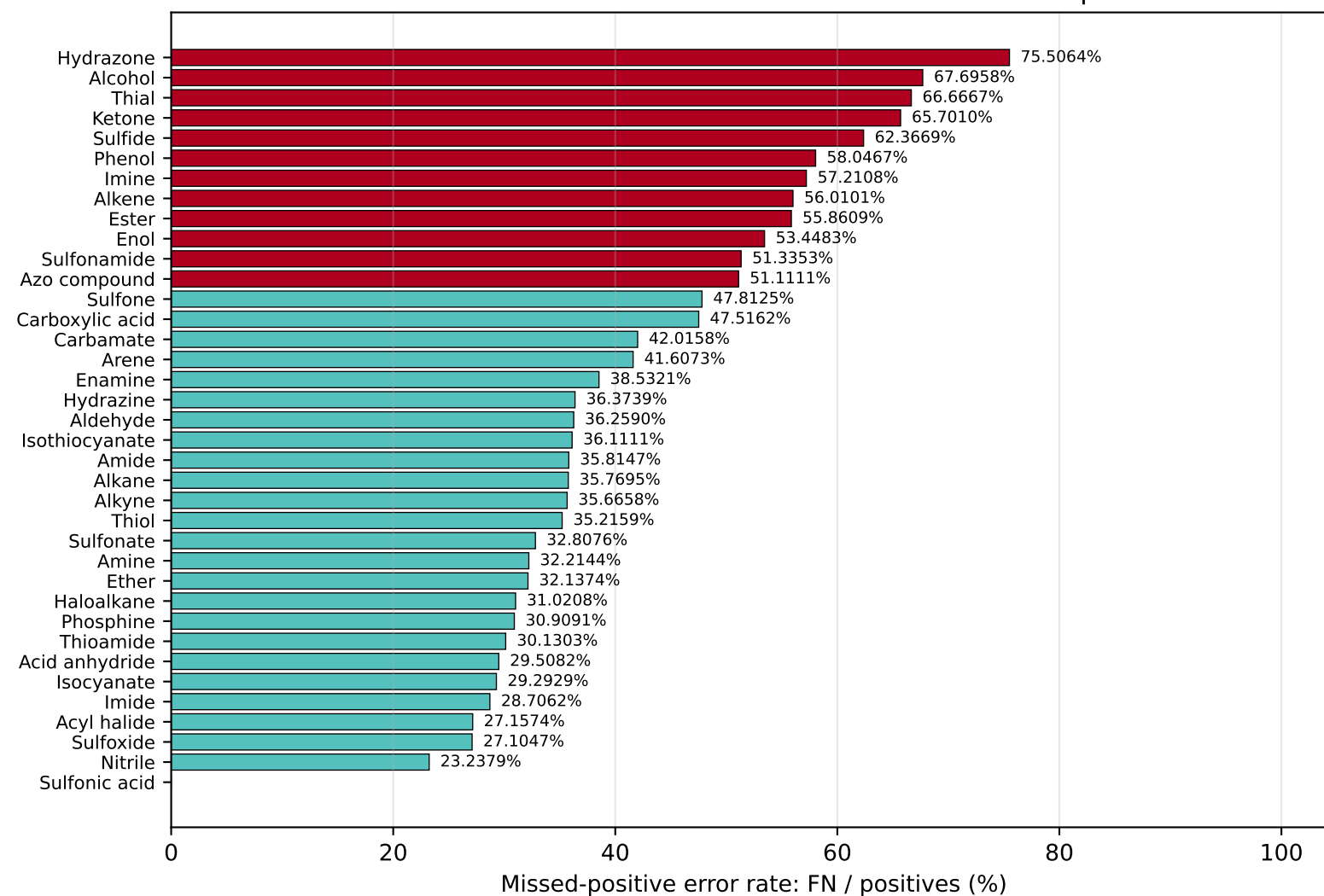
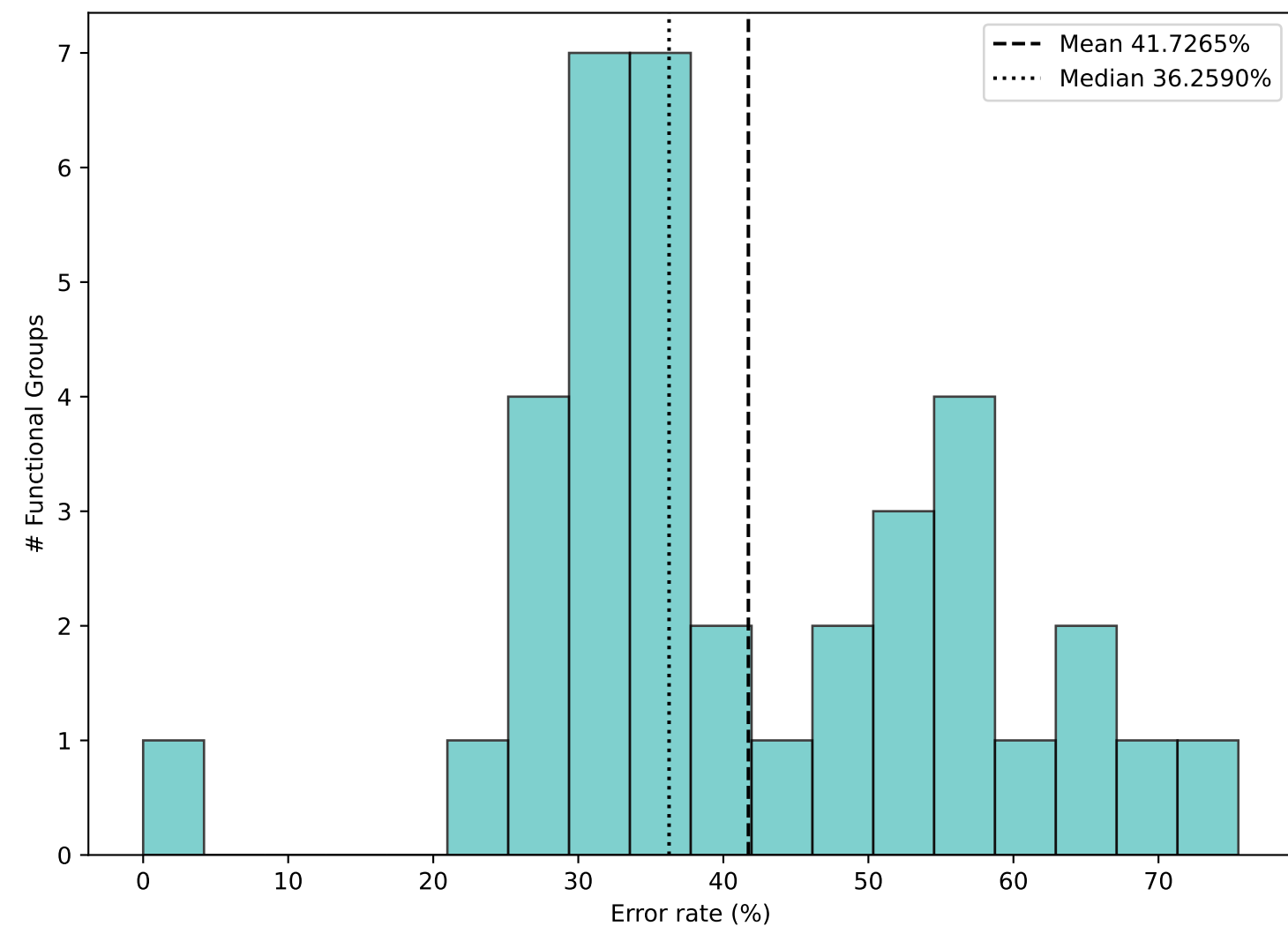


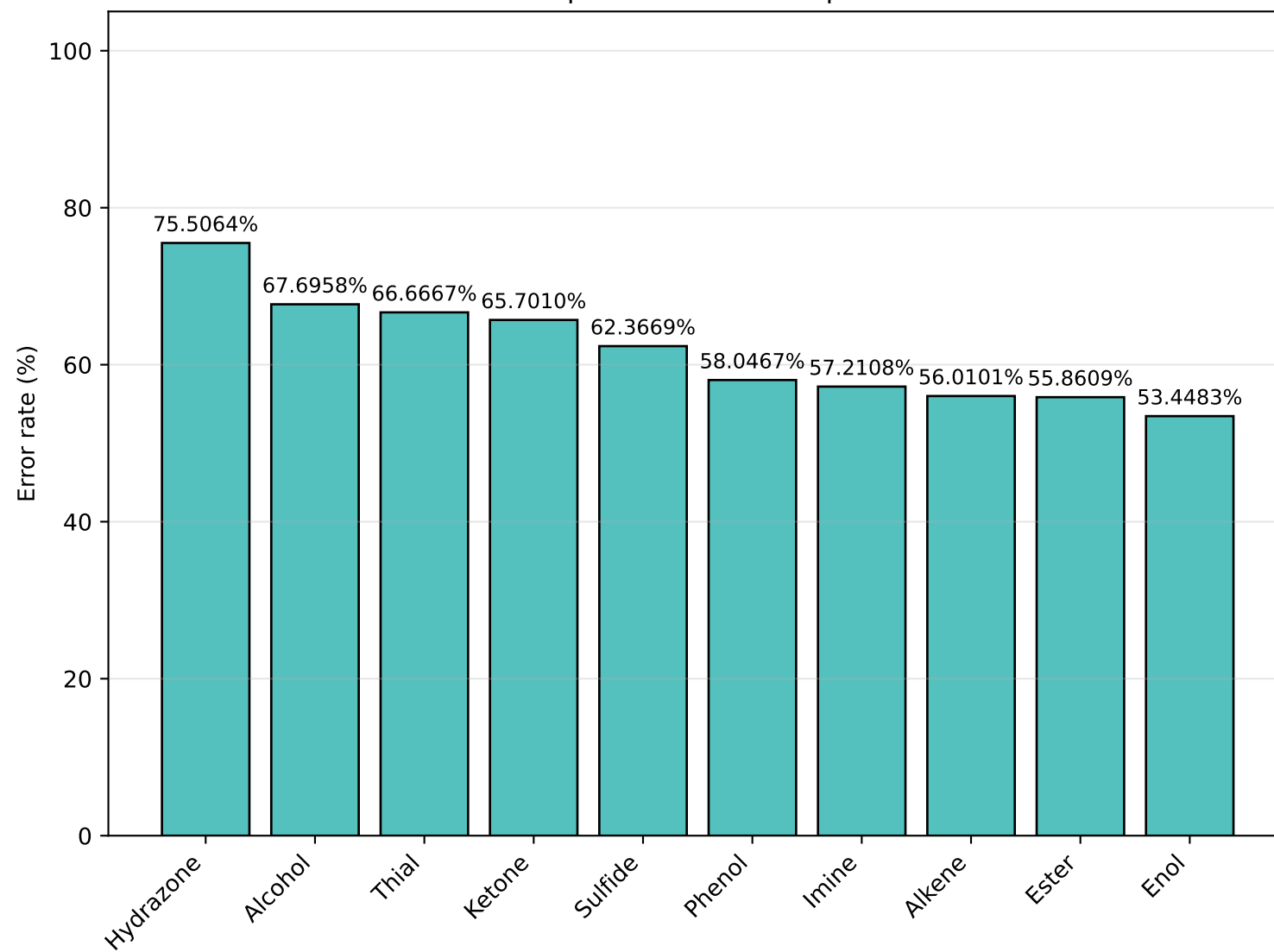
MPS MSMS- Validation Set - All 37 Functional Groups



Error-Rate Distribution



Top 10 Problem Groups



MPS MSMS- VALIDATION SET ERROR ANALYSIS

Global metrics:

F1 micro: 0.2854
 F1 macro: 0.1722
 Precision micro: 0.1878
 Recall micro: 0.5937
 Hamming accuracy: 0.6158

Error-rate statistics:

Mean: 41.7265%
 Median: 36.2590%
 Std: 15.4533%
 Max: 75.5064% (Hydrazone)
 Min: 0.0000% (Sulfonic acid)

Problem classes:

Severe (>50%): 12 groups
 High (20-50%): 24 groups
 Moderate (5-20%): 0 groups
 Low (<=5%): 1 groups